POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 2002B | Semester I / II | Practicum | 4.5 Credits

Engineering Physics for Applied Electrical Technology and Computing

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Applied electrical technology and computing programme group listed in the official syllabus**
Contact hours	**90**
Teaching scheme	**3:0:3:0**
Credits	**4.5**
CIE / ESE	**Theory 20 + Practical 20 / Theory 60**
Self-learning	**6 h/week; 90 h total**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2002B**, titled **Engineering Physics for Applied Electrical Technology and Computing**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

Source treatment: PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

Verified source note: No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

What You Are Expected to Learn

90
Contact hours

4.5
Credits

Theory 20 + Practical 20
CIE

Theory 60
ESE

Course introduction

Applied Physics develops physical reasoning, numerical discipline and laboratory skill for electrical technology and computing programmes. The syllabus links mechanics, properties of matter, rotational and wave motion, semiconductor physics, photoelectric effect and lasers to observation and measurement.

Malayalam support: ഈ handbook പഠിക്കുമ്പോൾ syllabus topic ആദ്യം വായിക്കുക; പിന്നീട് principle, diagram/procedure, application, common mistake എന്നിവ ക്രമമായി പഠിക്കുക.

Prerequisite foundation

Basic knowledge in Physics (Secondary Education).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	0	3	0	6 h/week; 90 h total	90	4.5	Theory 20 + Practical 20	Theory 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Develop broad understanding of physical principles and quantitative reasoning.
2. Build scientific temper for solving engineering problems.
3. Supplement factual knowledge through first-hand manipulation of apparatus.
4. Develop confidence in handling measuring equipment.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO എന്നത് exam-ൽ എന്ത് ചെയ്യാൻ കഴിയണം എന്ന് പറയുന്നു. Define, explain, apply എന്നീ action words ശ്രദ്ധിക്കുക.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Apply mechanics, including Newton's laws, work, energy and power, to motion-related problems and engineering applications.	Apply
CO2	Explain and apply properties of matter, fluid dynamics and thermodynamics to numerical problems and practical phenomena.	Apply
CO3	Analyze rotational motion and wave phenomena to solve engineering problems and interpret physical systems.	Apply
CO4	Explain and apply modern physics, including semiconductors, photoelectric effect and lasers, in engineering applications.	Apply

Official CO–PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Mechanics	Units; scalars and vectors; displacement, velocity and acceleration; Newton's laws; momentum; force; work, energy and power.	10 L hours + laboratory work	CO1	Module 1
2	Properties of Matter and Thermodynamics	Elasticity; fluid flow; Bernoulli principle; viscosity; heat and temperature; scales; heat transfer.	10 L hours + laboratory work	CO2	Module 2
3	Rotational and Wave Motion	Circular motion; angular quantities; moment of inertia; torque; angular momentum; SHM; transverse and longitudinal waves.	11 L hours + laboratory work	CO3	Module 3
4	Modern Physics	Energy bands; semiconductors; PN diode; photons; photoelectric effect; laser principles and ruby laser.	10 L hours + laboratory work	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Mechanics

Units; scalars and vectors; displacement, velocity and acceleration; Newton's laws; momentum; force; work, energy and power.

CO1 · 10 L hours + laboratory work

Properties of Matter and Thermodynamics

Elasticity; fluid flow; Bernoulli principle; viscosity; heat and temperature; scales; heat transfer.

CO2 · 10 L hours + laboratory work

Rotational and Wave Motion

Circular motion; angular quantities; moment of inertia; torque; angular momentum; SHM; transverse and longitudinal waves.

CO3 · 11 L hours + laboratory work

Modern Physics

Energy bands; semiconductors; PN diode; photons; photoelectric effect; laser principles and ruby laser.

CO4 · 10 L hours + laboratory work

MODULE 1

Mechanics

Units; scalars and vectors; displacement, velocity and acceleration; Newton’s laws; momentum; force; work, energy and power.

10 L hours + laboratory work **CO1** **Understand · Apply**

Learning goals

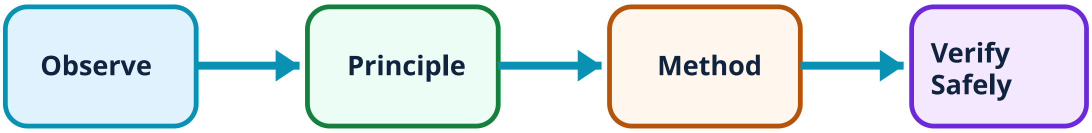
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Mechanics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Units, SI system and vectors–

A physical quantity is measured by comparing it with a standard. Fundamental quantities are independent base measurements, whereas derived quantities are constructed from them. SI units make laboratory results comparable. A scalar has magnitude only; a vector has magnitude and direction. Vector addition combines directional effects, while resolution replaces one vector by perpendicular components.

Study and application method

Write a quantity as value × unit; decide whether direction is relevant; draw the vector head-to-tail before using triangle or parallelogram law.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: The laboratory work includes verification of the parallelogram law of forces. Do not add magnitudes unless vectors are collinear.

2. Motion, Newton's laws and momentum–

Displacement describes change of position. Velocity is rate of change of displacement, and acceleration is rate of change of velocity. Newton's laws connect inertia, force and motion. Momentum is mass multiplied by velocity; conservation of momentum is useful when internal forces dominate during a short interaction such as recoil.

Study and application method

List known quantities, choose a sign convention and use $F = ma$ or $p = mv$ with SI units. In recoil, gun and projectile have equal and opposite momenta when external impulse is neglected.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Force and momentum are vectors; a negative answer normally indicates direction relative to the chosen positive direction.

3. Work, energy and power–

Work is energy transfer caused by a force through displacement. Kinetic energy is associated with motion and potential energy with position or configuration. Power is the rate at which work is done. Conservation of energy is a bookkeeping principle: energy changes form but is not created or destroyed in an isolated description.

Study and application method

Use $W = Fs$ when force and displacement are in the same direction, $KE = ½mv^2$, and $P = W/t$. Explain the unit joule or watt in the final answer.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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Common mistake / precaution: Electrical appliance power rating and electrical energy consumption use different quantities; keep watt and kWh distinct.

4. Mechanics laboratory practice–

The official experiments use vernier calipers, screw gauge, spherometer and the parallelogram-law apparatus. Each experiment develops a measurement chain: identify least count, remove zero error where applicable, take repeated readings, calculate the requested quantity and state the result with unit.

Study and application method

Prepare an observation table before readings. Keep the instrument clean, avoid excessive force on screw-gauge contacts and read scales with the eye normal to the scale.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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Common mistake / precaution: A result without unit, least-count awareness or safe handling is incomplete even if the arithmetic is correct.

Mechanical-work calculator

Force (N)

10

Displacement (m)

2

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Properties of Matter and Thermodynamics

Elasticity; fluid flow; Bernoulli principle; viscosity; heat and temperature; scales; heat transfer.

10 L hours + laboratory work | CO2 | Understand · Apply

Learning goals

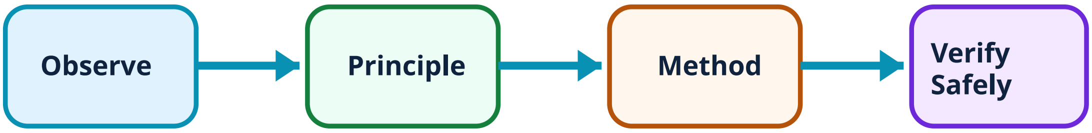
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Properties of Matter and Thermodynamics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Elasticity, stress, strain and Hooke’s law–

Elasticity is the ability of a material to regain original shape after removal of a deforming force within its elastic limit. Stress is internal restoring force per unit area, and strain is fractional deformation. Hooke’s law states that stress is proportional to strain within elastic limit. Young’s, rigidity and bulk moduli describe different forms of deformation.

Study and application method

Identify deformation type, then select relevant modulus. For Young’s-modulus problems, convert lengths and areas consistently to SI units before substitution.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Hooke's law does not state that every material remains proportional for every load; elastic limit is essential.

2. Fluid dynamics, continuity, Bernoulli principle and viscosity–

Streamline flow has an orderly path; turbulent flow has irregular mixing. For steady incompressible flow, continuity expresses conservation of volume flow rate. Bernoulli's principle relates pressure, speed and height along a streamline. Viscosity is internal resistance to relative motion of fluid layers; Stokes' law describes viscous force on a small sphere moving slowly through a viscous fluid.

Study and application method

Use a labelled flow diagram and state assumptions before applying a relation. Explain an atomizer through a fast air stream associated with lower pressure near the liquid tube.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
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Common mistake / precaution: Do not apply Bernoulli’s principle unchanged to every real, turbulent or highly viscous flow.

3. Heat, temperature and heat transfer–

Temperature indicates thermal state; heat is energy transferred because of temperature difference. Celsius, Fahrenheit and Kelvin scales are related by conversion. Conduction transfers energy through a body, convection uses bulk fluid motion and radiation transfers energy by electromagnetic waves.

Study and application method

Distinguish the measured quantity from the process of transfer. In conversions, retain scale symbol and check whether the answer is physically reasonable.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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Common mistake / precaution: A thermometer measures its own equilibrium temperature; allow sufficient time and avoid treating the instrument as a heat source.

4. Properties-of-matter laboratory practice–

The official work includes a mercury thermometer, Hooke's-law apparatus, Stokes' law and U-tube experiments. These relate carefully observed dimensions, time or level difference to a physical quantity.

Study and application method

Take repeated readings, control one variable at a time, tabulate units and identify possible reading error before calculating the final result.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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Common mistake / precaution: Glassware, liquids and weights require stable setup and teacher-approved handling.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Rotational and Wave Motion

Circular motion; angular quantities; moment of inertia; torque; angular momentum; SHM; transverse and longitudinal waves.

11 L hours + laboratory work | CO3 | Understand · Apply

Learning goals

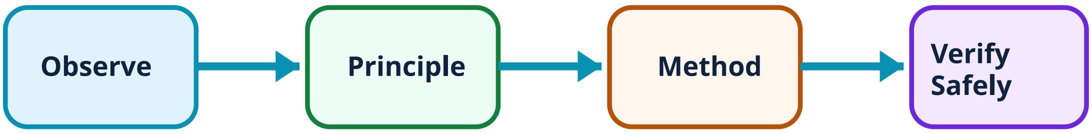
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Rotational and Wave Motion: identify the system, state its principle, follow the correct method and verify the result safely.

1. Circular and rotational motion–

Angular displacement, angular velocity and angular acceleration describe rotation. Linear and angular quantities are related through radius. Centripetal acceleration is directed toward the centre of a circular path, and centripetal force is the net inward force needed for circular motion.

Study and application method

Draw radius and inward direction before resolving forces. Use $v = r\omega$ and $a_t = r\alpha$ only with consistent units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
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Common mistake / precaution: Centripetal force is not an additional outward force; it names the resultant inward force.

2. Moment of inertia, torque and angular momentum–

Moment of inertia measures resistance to angular acceleration and depends on mass distribution about the axis. Torque is turning effect of a force. Parallel- and perpendicular-axis theorems relate moments about axes. Angular momentum is conserved when external torque is negligible.

Study and application method

Identify axis first. Select stated formula for rod, disc, ring or sphere and use a clear diagram for torque.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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Common mistake / precaution: Mass alone does not determine rotational inertia; distance from axis matters.

3. SHM and waves–

Periodic motion repeats after equal intervals. Simple harmonic motion has restoring tendency toward equilibrium in the ideal model. Waves transfer disturbance and energy; transverse and longitudinal waves differ in particle-vibration direction. Wave speed, frequency and wavelength are related by $v = f\lambda$.

Study and application method

Write all wave quantities with units, convert prefixes correctly and solve only after selecting the relation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Frequency is cycles per second; it is not the same quantity as wave speed.

4. Rotational and wave laboratory practice–

The official practical work includes simple pendulum, moment bar, spring-oscillation, resonance-column and flywheel experiments. Good practice requires repeated observations, controlled variables, a graph or calculation path and a result linked to aim.

Study and application method

Check alignment, avoid parallax, take multiple timing cycles for periodic motion and record relevant conditions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Never use damaged strings, unstable flywheel setups or unsupported masses.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Modern Physics

Energy bands; semiconductors; PN diode; photons; photoelectric effect; laser principles and ruby laser.

10 L hours + laboratory work | CO4 | Understand · Apply

Learning goals

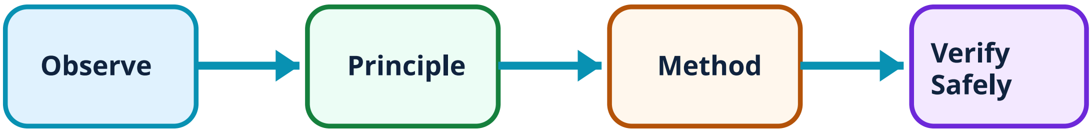
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Modern Physics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Energy bands and semiconductor diode–

Energy-band structure explains conductor, insulator and semiconductor behaviour. Intrinsic semiconductors are pure; doping produces extrinsic material. A PN junction has a depletion region and different forward/reverse-bias behaviour. Its V–I characteristic relates current to applied voltage.

Study and application method

Begin with material type and junction direction. For a diode graph, label axes, polarity and knee region.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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Common mistake / precaution: Doping changes carrier concentration; it is not the same as charging an object.

2. Photon and photoelectric effect–

Quantum theory treats light as packets called photons. Photon energy depends on frequency. Photoelectric emission occurs only when incident frequency reaches threshold frequency; Einstein's equation expresses energy balance between photon energy, work function and electron kinetic energy.

Study and application method

State given quantities, use consistent SI units and distinguish roles of intensity and frequency in the ideal explanation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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Common mistake / precaution: Increasing intensity below threshold frequency does not supply missing photon energy in the simple model.

3. Lasers and ruby laser–

Laser action uses stimulated emission, population inversion and an optical resonator to produce a directional, coherent and nearly monochromatic beam. Ruby laser is the official example of construction and working.

Study and application method

Use an energy-level sketch to distinguish absorption, spontaneous emission and stimulated emission, then list applications after explaining the principle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
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Common mistake / precaution: Never view a laser beam directly or point it toward reflective surfaces.

4. Modern-physics laboratory practice–

The syllabus includes plotting V–I characteristics of Ge/Si diodes and studying a photoelectric cell. Work must connect circuit arrangement, controlled variable, observed response and graph interpretation.

Study and application method

Verify supply settings before connection, use current-limiting resistance and record readings with correct units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
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Common mistake / precaution: Switch off supply before changing connections and do not exceed device ratings.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Mechanics

Use the concept flow to revise observation, principle, method and verification.

Module 2: Properties of Matter and Thermodynamics

Use the concept flow to revise observation, principle, method and verification.

Module 3: Rotational and Wave Motion

Use the concept flow to revise observation, principle, method and verification.

Module 4: Modern Physics

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Classify quantities and state SI units; practise vector addition and resolution.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Solve motion, force, momentum, work, energy and power problems.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create concept maps and home observations for elasticity, fluid flow, viscosity and heat transfer.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Complete the official modern-physics activities on semiconductors, diode behaviour, photoelectric effect and laser principles.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	Regular attendance and punctuality.
Continuous evaluation	Theory 20 + Practical 20	Preparation, setup, observation, analysis, viva, record and discipline.
Practical tests	As stated in supplied PDF	Procedure/write-up, execution, result, viva and certified record.
Open-ended experiment	Where listed in supplied PDF	Originality, plan, execution/data, analysis/presentation and teamwork.
End-semester assessment	Theory 60	Safe completion, record and viva according to official scheme.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practicaltest scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Mechanics

Explain Units, SI system and vectors and state one correct method, application or precaution.–

A physical quantity is measured by comparing it with a standard. Fundamental quantities are independent base measurements, whereas derived quantities are constructed from them. SI units make laboratory results comparable. A scalar has magnitude only; a vector has magnitude and direction. Vector addition combines directional effects, while resolution replaces one vector by perpendicular components.

Answer framework: Write a quantity as value × unit; decide whether direction is relevant; draw the vector head-to-tail before using triangle or parallelogram law.

Important point: The laboratory work includes verification of the parallelogram law of forces. Do not add magnitudes unless vectors are collinear.

Explain Motion, Newton's laws and momentum and state one correct method, application or precaution.–

Displacement describes change of position. Velocity is rate of change of displacement, and acceleration is rate of change of velocity. Newton's laws connect inertia, force and motion. Momentum is mass multiplied by velocity; conservation of momentum is useful when internal forces dominate during a short interaction such as recoil.

Answer framework: List known quantities, choose a sign convention and use $F = ma$ or $p = mv$ with SI units. In recoil, gun and projectile have equal and opposite momenta when external impulse is neglected.

Important point: Force and momentum are vectors; a negative answer normally indicates direction relative to the chosen positive direction.

Explain Work, energy and power and state one correct method, application or precaution.–

Work is energy transfer caused by a force through displacement. Kinetic energy is associated with motion and potential energy with position or configuration. Power is the rate at which work is done. Conservation of energy is a bookkeeping principle: energy changes form but is not created or destroyed in an isolated description.

Answer framework: Use W = Fs when force and displacement are in the same direction, KE = $\frac{1}{2}mv^2$, and P = W/t. Explain the unit joule or watt in the final answer.

Important point: Electrical appliance power rating and electrical energy consumption use different quantities; keep watt and kWh distinct.

Explain Mechanics laboratory practice and state one correct method, application or precaution.–

The official experiments use vernier calipers, screw gauge, spherometer and the parallelogram-law apparatus. Each experiment develops a measurement chain: identify least count, remove zero error where applicable, take repeated readings, calculate the requested quantity and state the result with unit.

Answer framework: Prepare an observation table before readings. Keep the instrument clean, avoid excessive force on screw-gauge contacts and read scales with the eye normal to the scale.

Important point: A result without unit, least-count awareness or safe handling is incomplete even if the arithmetic is correct.

Module 2 — Properties of Matter and Thermodynamics

Explain Elasticity, stress, strain and Hooke's law and state one correct method, application or precaution.–

Elasticity is the ability of a material to regain original shape after removal of a deforming force within its elastic limit. Stress is internal restoring force per unit area, and strain is fractional deformation. Hooke's law states that stress is proportional to strain within elastic limit. Young's, rigidity and bulk moduli describe different forms of deformation.

Answer framework: Identify deformation type, then select relevant modulus. For Young's-modulus problems, convert lengths and areas consistently to SI units before substitution.

Important point: Hooke's law does not state that every material remains proportional for every load; elastic limit is essential.

Explain Fluid dynamics, continuity, Bernoulli principle and viscosity and state one correct method, application or precaution.–

Streamline flow has an orderly path; turbulent flow has irregular mixing. For steady incompressible flow, continuity expresses conservation of volume flow rate. Bernoulli's principle relates pressure, speed and height along a streamline. Viscosity is internal resistance to relative motion of fluid layers; Stokes' law describes viscous force on a small sphere moving slowly through a viscous fluid.

Answer framework: Use a labelled flow diagram and state assumptions before applying a relation. Explain an atomizer through a fast air stream associated with lower pressure near the liquid tube.

Important point: Do not apply Bernoulli's principle unchanged to every real, turbulent or highly viscous flow.

Explain Heat, temperature and heat transfer and state one correct method, application or precaution.–

Temperature indicates thermal state; heat is energy transferred because of temperature difference. Celsius, Fahrenheit and Kelvin scales are related by conversion. Conduction transfers energy through a body, convection uses bulk fluid motion and radiation transfers energy by electromagnetic waves.

Answer framework: Distinguish the measured quantity from the process of transfer. In conversions, retain scale symbol and check whether the answer is physically reasonable.

Important point: A thermometer measures its own equilibrium temperature; allow sufficient time and avoid treating the instrument as a heat source.

Explain Properties-of-matter laboratory practice and state one correct method, application or precaution.–

The official work includes a mercury thermometer, Hooke's-law apparatus, Stokes' law and U-tube experiments. These relate carefully observed dimensions, time or level difference to a physical quantity.

Answer framework: Take repeated readings, control one variable at a time, tabulate units and identify possible reading error before calculating the final result.

Important point: Glassware, liquids and weights require stable setup and teacher-approved handling.

Module 3 — Rotational and Wave Motion

Explain Circular and rotational motion and state one correct method, application or precaution.–

Angular displacement, angular velocity and angular acceleration describe rotation. Linear and angular quantities are related through radius. Centripetal acceleration is directed toward the centre of a circular path, and centripetal force is the net inward force needed for circular motion.

Answer framework: Draw radius and inward direction before resolving forces. Use $v = r\omega$ and $a_t = r\alpha$ only with consistent units.

Important point: Centripetal force is not an additional outward force; it names the resultant inward force.

Explain Moment of inertia, torque and angular momentum and state one correct method, application or precaution.–

Moment of inertia measures resistance to angular acceleration and depends on mass distribution about the axis. Torque is turning effect of a force. Parallel- and perpendicular-axis theorems relate moments about axes. Angular momentum is conserved when external torque is negligible.

Answer framework: Identify axis first. Select stated formula for rod, disc, ring or sphere and use a clear diagram for torque.

Important point: Mass alone does not determine rotational inertia; distance from axis matters.

Explain SHM and waves and state one correct method, application or precaution.–

Periodic motion repeats after equal intervals. Simple harmonic motion has restoring tendency toward equilibrium in the ideal model. Waves transfer disturbance and energy; transverse and longitudinal waves differ in particle-vibration direction. Wave speed, frequency and wavelength are related by $v = f\lambda$.

Answer framework: Write all wave quantities with units, convert prefixes correctly and solve only after selecting the relation.

Important point: Frequency is cycles per second; it is not the same quantity as wave speed.

Explain Rotational and wave laboratory practice and state one correct method, application or precaution.–

The official practical work includes simple pendulum, moment bar, spring-oscillation, resonance-column and flywheel experiments. Good practice requires repeated observations, controlled variables, a graph or calculation path and a result linked to aim.

Answer framework: Check alignment, avoid parallax, take multiple timing cycles for periodic motion and record relevant conditions.

Important point: Never use damaged strings, unstable flywheel setups or unsupported masses.

Module 4 — Modern Physics

Explain Energy bands and semiconductor diode and state one correct method, application or precaution.–

Energy-band structure explains conductor, insulator and semiconductor behaviour. Intrinsic semiconductors are pure; doping produces extrinsic material. A PN junction has a depletion region and different forward/reverse-bias behaviour. Its V–I characteristic relates current to applied voltage.

Answer framework: Begin with material type and junction direction. For a diode graph, label axes, polarity and knee region.

Important point: Doping changes carrier concentration; it is not the same as charging an object.

Explain Photon and photoelectric effect and state one correct method, application or precaution.–

Quantum theory treats light as packets called photons. Photon energy depends on frequency. Photoelectric emission occurs only when incident frequency reaches threshold frequency; Einstein’s equation expresses energy balance between photon energy, work function and electron kinetic energy.

Answer framework: State given quantities, use consistent SI units and distinguish roles of intensity and frequency in the ideal explanation.

Important point: Increasing intensity below threshold frequency does not supply missing photon energy in the simple model.

Explain Lasers and ruby laser and state one correct method, application or precaution.–

Laser action uses stimulated emission, population inversion and an optical resonator to produce a directional, coherent and nearly monochromatic beam. Ruby laser is the official example of construction and working.

Answer framework: Use an energy-level sketch to distinguish absorption, spontaneous emission and stimulated emission, then list applications after explaining the principle.

Important point: Never view a laser beam directly or point it toward reflective surfaces.

Explain Modern-physics laboratory practice and state one correct method, application or precaution.–

The syllabus includes plotting V–I characteristics of Ge/Si diodes and studying a photoelectric cell. Work must connect circuit arrangement, controlled variable, observed response and graph interpretation.

Answer framework: Verify supply settings before connection, use current-limiting resistance and record readings with correct units.

Important point: Switch off supply before changing connections and do not exceed device ratings.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Mechanics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Properties of Matter and Thermodynamics.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Rotational and Wave Motion.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Modern Physics.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Units; scalars and vectors; displacement, velocity and acceleration; Newton's laws; momentum; force; work, energy and power..
2. Develop a complete answer or practical plan for: Elasticity; fluid flow; Bernoulli principle; viscosity; heat and temperature; scales; heat transfer..
3. Develop a complete answer or practical plan for: Circular motion; angular quantities; moment of inertia; torque; angular momentum; SHM; transverse and longitudinal waves..
4. Develop a complete answer or practical plan for: Energy bands; semiconductors; PN diode; photons; photoelectric effect; laser principles and ruby laser..

LAST-MILE REVISION

Quick Revision

Module 1: Mechanics

Units; scalars and vectors; displacement, velocity and acceleration; Newton's laws; momentum; force; work, energy and power.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Properties of Matter and Thermodynamics

Elasticity; fluid flow; Bernoulli principle; viscosity; heat and temperature; scales; heat transfer.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Rotational and Wave Motion

Circular motion; angular quantities; moment of inertia; torque; angular momentum; SHM; transverse and longitudinal waves.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Modern Physics

Energy bands; semiconductors; PN diode; photons; photoelectric effect; laser principles and ruby laser.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Units, SI system and vectors	A physical quantity is measured by comparing it with a standard.
Motion, Newton's laws and momentum	Displacement describes change of position.
Work, energy and power	Work is energy transfer caused by a force through displacement.
Mechanics laboratory practice	The official experiments use vernier calipers, screw gauge, spherometer and the parallelogram-law apparatus.
Elasticity, stress, strain and Hooke's law	Elasticity is the ability of a material to regain original shape after removal of a deforming force within its elastic limit.
Fluid dynamics, continuity, Bernoulli principle and viscosity	Streamline flow has an orderly path; turbulent flow has irregular mixing.
Heat, temperature and heat transfer	Temperature indicates thermal state; heat is energy transferred because of temperature difference.
Properties-of-matter laboratory practice	The official work includes a mercury thermometer, Hooke's-law apparatus, Stokes' law and U-tube experiments.
Circular and rotational motion	Angular displacement, angular velocity and angular acceleration describe rotation.
Moment of inertia, torque and angular momentum	Moment of inertia measures resistance to angular acceleration and depends on mass distribution about the axis.
SHM and waves	Periodic motion repeats after equal intervals.
Rotational and wave laboratory practice	The official practical work includes simple pendulum, moment bar, spring-oscillation, resonance-column and flywheel experiments.
Energy bands and semiconductor diode	Energy-band structure explains conductor, insulator and semiconductor behaviour.
Photon and photoelectric effect	Quantum theory treats light as packets called photons.
Lasers and ruby laser	Laser action uses stimulated emission, population inversion and an optical resonator to produce a directional, coherent and nearly monochromatic beam.
Modern-physics laboratory practice	The syllabus includes plotting V–I characteristics of Ge/Si diodes and studying a photoelectric cell.

LEARNING RESOURCES

References

Officially listed print resources

1. NCERT, Text Book of Physics for Class XI & XII (Part I and II).
2. Dr. H. H. Lal, Applied Physics, Vol. I and Vol. II.

3. Halliday, Resnick and Walker, Fundamentals of Physics.

Officially listed web resources

- https://phydemo.app
- https://phet.colorado.edu/en

Links are reproduced because they appear in the supplied syllabus. Use institutional safety and access guidance when opening external resources.